

Temporal Foundation–Model Detection of River–Ice Onset and Breakup

Interim Discussion: Results to Date, and Predictions for the Next Phases

FM_{Ice} project — working note

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Summary

We detect river–ice *onset* and *breakup* from time-lapse camera imagery with a light temporal head trained on *frozen* video foundation–model (V-JEPA 2) embeddings, benchmarked against an image foundation model (DINOv2) and a published per-frame threshold baseline (Rice-Net). The contribution is a measured result — event-timing error — not an architecture. Status: data, embeddings, and the static separability gate are complete; the temporal head (Phase 3) is trained and evaluated; the change-point baselines, transfer test, and evaluation/figures code (Phases 4–6) are written and pending experiments.

Headline so far. On a leave-one-winter-out split of the training station (Cedarburg), the temporal head reads *breakup* to within **2–22 h** of the USGS ice flag, and *onset* to within **18 h** on one winter. Onset on the other winter is degraded by an early-season (November) appearance confound; a physically-motivated freezing guard cuts that error roughly in half. The best configuration (V-JEPA, TCN, freezing guard) averages **122 h** timing error and beats both the per-frame anchor (494 h) and DINOv2 (191 h) — a *tentative positive* signal for the video model on the timing axis, opposite to the static-separability result below. All numbers are from $n = 2$ winters and are directional.

1 Problem and hypotheses

The encoder is frozen; only a small temporal head trains, on CPU. Four hypotheses drive the work:

- H1** Temporal modeling on FM embeddings beats per-frame thresholding on timing error.
- H2** Video-FM (V-JEPA) beats image-FM (DINOv2) for this task.
- H3** FM embeddings separate ice from open water across lighting better than pixels.
- H4** (optional) A forecast head gives non-trivial lead time before onset.

2 Data and evaluation protocol

Two stations, hourly overlay imagery, with USGS daily ice flags as the solid reference and a stage (gage-height) change-point series as a rough cross-check. A *clip* is 16 hourly frames, 4h stride (frozen after embeddings were cached). Per clip we cache one 1024-d embedding for each encoder.

Station	Winter(s)	Clips (ice%)	Role
Cedarburg	2022–2023	1099 (24%)	train
Cedarburg	2023–2024	1173 (13%)	train
Bismarck	2024–2025	1130 (52%)	transfer test (held out)

Split deviation (deliberate). The plan specifies holding out winter 2024–2025 on the train station, but Cedarburg has only two winters and 2024–2025 belongs to Bismarck, the reserved transfer station. We therefore evaluate the train station by *leave-one-winter-out* on Cedarburg and touch Bismarck only in the Phase 5 transfer test. With two winters there is no separate validation split, so the head trains for a fixed number of epochs with config hyperparameters and no test peeking.

3 Results to date

3.1 H3 — GATE A: static ice/water separability (PASS, both encoders)

A linear probe on frozen embeddings, scored out-of-fold with time-blocked cross-validation (clips overlap, so random folds would leak), separates ice from open water strongly and the separation survives lighting changes (Table 1). Both encoders PASS. Critically, *H2 is negative at this gate*: DINOv2 ties or slightly beats V-JEPA on pooled and especially night AUC. Reading: *per-clip* ice/water discrimination is an appearance problem, not a motion problem — which is exactly what an image FM is built for.

Table 1: GATE A pooled / night ROC-AUC of a linear probe (out-of-fold, time-blocked). “light-only” is a night-fraction+brightness baseline: the embedding margin over it is the honest signal.

Station-winter	V-JEPA pooled/night	DINOv2 pooled/night	light-only
Cedarburg 2022–2023	0.927 / 0.909	0.973 / 0.974	0.868
Cedarburg 2023–2024	0.983 / 0.944	0.973 / 0.946	0.893
Bismarck 2024–2025	0.946 / 0.920	0.959 / 0.918	0.703

3.2 H1 — the temporal head (Phase 3)

The head (default: MS-TCN-style dilated temporal conv, class-balanced BCE + smoothing loss) emits a per-step ice-state probability; onset and breakup are read off as the first and last *sustained* ice runs, mirroring the reference’s run-length rule. Table 2 gives timing error vs the USGS ice flag.

Table 2: Onset / breakup timing error in hours vs the USGS ice flag (leave-one-winter-out on Cedarburg; lower is better). “+guard” applies the physical freezing guard.

Model	2022–2023		2023–2024		mean
	onset	breakup	onset	breakup	
V-JEPA TCN +guard	446	2	18	22	122
V-JEPA TCN (no guard)	1146	2	18	22	297
DINOv2 TCN	710	2	22	30	191
V-JEPA per-frame (anchor)	726	70	1122	58	494

[Figure *fig_timing.png*: onset vs breakup timing error, mean over the two held-out winters, log scale — saved in *results/figures/discussion/*.]

Two robust observations. (1) **Breakup is essentially solved** (2–22 h) for the temporal head: spring ice-out is an abrupt, well-lit appearance change. (2) **Onset is bimodal**. On 2023–2024 the head nails it (18 h); on 2022–2023 it fires a false “ice” block in November, weeks before the real December onset. This appears for *both* encoders and even for the linear per-frame probe, so it is a feature/label-transfer confound — November scenes (low sun, leaf-off, dark water) look ice-like to a probe trained on the other winter — not merely head overfitting.

The freezing guard. River ice does not persist while the air is well above freezing, so a sustained ice run only counts if its mean air temperature is at or below 0°C — a domain prior, not a fit-to-test threshold. It moves the 2022–2023 predicted onset from Nov 1 to Nov 30 (reference: Dec 19), cutting that error from 1146 h to 446 h and making V-JEPA+TCN+guard the best model overall.

[Figure *fig_nov_guard.png*: the head’s $P(\text{ice})$ rises in warm November before the true December ice season; the guard rejects the warm-mean run and shifts the predicted onset from Nov 1 to Nov 30, within ~ 19 days of the Dec 19 reference — saved in *results/figures/discussion/*.]

H1 verdict (interim). For V-JEPA the temporal head clearly beats its per-frame anchor (122 vs 494 h mean; on 2023–2024 onset, 18 vs 1122 h — the head catches a brief late-November freeze the per-frame probe misses entirely). H1 is *supported but fragile* at $n = 2$.

3.3 H2 — V-JEPA vs DINOv2 on timing

On the timing axis the best V-JEPA head (122 h) edges DINOv2 (191 h), driven by the guard’s larger effect on V-JEPA’s November confound. This is a *tentative reversal* of the negative static GATE-A result: motion/temporal context may help where per-clip appearance does not. It is not decisive — the gap rests on one winter and one post-hoc guard. The clean H2 comparison is Phase 4 (identical head, identical splits, plus change-point and RIce-Net rows) and GATE B.

3.4 RIce-Net anchor — paused

The published RIce-Net weights are downloaded and the pipeline is wired, but the model ships as a *pickled full object* (arbitrary-code execution on load) and is incompatible with the installed *segmentation-models-pytorch* (decoder internals changed). Running it was paused pending a decision on (a) authorizing the unpickle of the official artifact and (b) pinning the library version vs. rebuilding from the state dict. Until then, the per-frame probe serves as the available H1 anchor.

4 Discussion

Why breakup is easy and onset is hard. Breakup is a sharp, high-contrast, well-lit transition (ice $\rightarrow$ moving open water); both the appearance signal and the label are crisp. Onset is gradual (frazil, skim ice, freeze-thaw stutter), happens in the darkest, lowest-sun part of the year, and its label (a daily flag) is coarse relative to the real physical transition. The dominant error mode is therefore a *warm-season appearance confound at onset*, which is why a physical temperature prior — not a bigger model — is the right lever.

The H2 tension is the interesting result. Static separability favors the image model (DINOv2); timing, tentatively, favors the video model (V-JEPA). If this holds with proper baselines, the

project’s thesis sharpens: video-FM’s value here is *temporal*, not per-frame, and the honest way to show it is the event-timing table, not a separability AUC.

Core limitation: label scarcity. Two training winters means one onset and one breakup event each; every number is directional, and a single confounded November dominates the onset average. This is the project’s named risk, and it is why the design leans on frozen features, a light head, label-free change-point baselines, and a physical prior.

5 Next steps and predictions

Phases 4–6 code is written and import-clean; the remaining work is to run it.

Table 3: Predictions for the next phases (with rough confidence).

Step	Prediction	Conf.
RIce-Net baseline (resolve pickle)	Temporal head beats RIce-Net on breakup clearly; competitive on onset. RIce-Net’s own threshold will also stumble on the warm-November confound.	med
Phase 4: change-point + GATE B	BOCPD/BEAST on the embedding stream find breakup well, onset poorly (same confound, no temperature prior). GATE B: V-JEPA $\geq$ DINOv2 on timing but within noise $\Rightarrow$ reframe as “temporal beats threshold.”	med
Phase 5: transfer to Bismarck	Breakup transfers (small gap); onset gap larger. Bismarck is <i>icier</i> (52%) and colder, so the freezing guard helps more and pooled timing may actually be competitive despite zero Bismarck training.	med-low
Onset improvement	The freezing guard + a third winter (or station pooling) is worth more than any head change. A learned temperature gate or a forecast head (H4) is the natural extension.	med

Concrete actions, in order. (1) Decide the RIce-Net unpickle/version question and run the anchor. (2) Run Phase 4 change-point baselines and assemble the single H2 table; call GATE B. (3) Run the Phase 5 transfer test and report the gap honestly. (4) Generate the Phase 6 metrics, error-analysis, and H3/timeline figures. (5) Only then consider H4 (forecast lead time), and only if it does not threaten the MVP freeze.

Phase 5 transfer test (measured, held-out station)

The temporal head, per-frame probe, and feature scaler are fit on *both* Cedarburg winters only; Bismarck 2024–2025 is transformed and predicted with no retraining and no statistics fit on it (no leakage). Table 4 reports timing error vs the USGS ice flag, with the Cedarburg leave-one-winter-out (LOO) mean as the anchor and the gap defined as Bismarck minus that anchor.

What actually happened (vs the prediction in Table 3). The prediction was *inverted*. We expected breakup to transfer with a small gap and onset to show the larger gap; instead *onset* transfers essentially perfectly (18–22 h on a station the model never saw) while *breakup* is the harder transfer (122–150 h, vs 2–22 h on Cedarburg). The head also beats the per-frame probe on Bismarck onset (18 h vs 306 h), so H1 generalizes off-station.

Table 4: Phase 5 transfer to Bismarck (no retraining). Timing error in hours vs the USGS ice flag; lower is better.

Config	Bismarck onset	Bismarck breakup	Cedarburg LOO mean	Gap (mean)
V-JEPA / TCN / +guard	18	122	122	−52
V-JEPA / TCN (no guard)	18	122	297	−227
DINOv2 / TCN ($\pm$ guard)	22	150	191	−105

The freezing guard is Cedarburg-specific. Bismarck onset is 18 h with or without the guard: the icier (52%), colder station has no warm-November false positive for the guard to remove. The guard only moves the Cedarburg anchor (122 vs 297 h), which is why the V-JEPA gap differs by row. The negative gaps are real but partly an artifact of the Nov-2022 confound inflating the Cedarburg onset average — the absolute Bismarck errors are the honest headline.

A cleaner H2 than GATE B. On the held-out station V-JEPA beats DINOv2 on both events (onset 18 vs 22 h, breakup 122 vs 150 h; per-frame AUC 0.99 vs 0.98), and, unlike the guard-dependent Cedarburg GATE B, this does *not* rest on the freezing guard. The transfer test gives the project its clearest video-FM > image-FM evidence. With Phase 5 complete, the MVP is frozen.

Honest caveats

$n = 2$ winters; no validation split; the freezing guard is a sensible prior but its 0°C threshold is a choice; the H2 timing edge rests on one winter and one guard; the stage change-point reference is crude (it mis-dates a late-February freshet as onset) and is used only as a cross-check. None of these are hidden in the averages above — they *are* the averages, and the per-winter, per-event numbers are reported alongside.